

OCTA RESEARCH

Technical Note

Shadow Math / Calculus

Projection–Lift Geometry and Audit-Grade Dimension Certificates

The Environment becomes your weapon.

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Abstract

Purpose. “Shadow Math” is a projection-first formalism: observed lower-dimensional shadows constrain (and sometimes uniquely determine) higher-dimensional structure through lifts and consistency. We present three core theorems:

- (A) **Single-view non-injectivity:** a single 2D projection does not uniquely determine 3D.
- (B) **Multi-view lift:** with two calibrated non-degenerate views and correct correspondences, a 3D point is uniquely recovered by triangulation.
- (C) **Dimension certificate:** from pairwise distances, the Gram (double-centered) matrix eigenstructure certifies the minimal Euclidean embedding dimension.

Empirical evidence. We embed OCTA Proof Lab v1.3 (JSON) and summarize its audit-grade certificate: a 3D world is non-planar while a 2D-in-3D control is planar under noise, using eigen-ratios, out-of-plane energy, and MDS stress.

Contents

1	Shadow Math: Core Definitions	3
2	Theorem A: Single-View Non-Injectivity	3
2.1	Pinhole projection model	3
3	Theorem B: Multi-View Lift via Triangulation	3
3.1	Back-projected rays	3
4	Shadow Calculus: Fiber Dimension from Jacobian Rank	4

5	Theorem C: Gram Eigenstructure Certifies Dimension	4
5.1	Distances, centering, and Gram	4
5.2	Noise-robust planarity certificate	5
6	Empirical Evidence: Proof Lab v1.3 (Audit Embed)	5
6.1	Experiment summary	5
6.2	Key outcomes (v1.3)	6
6.3	Audit table	6
7	Interpretation: What is Proven and What is Not	6
7.1	What is proven (mathematically)	6
7.2	What is not proven (physics)	6
A	Appendix A: Embedded Audit Artifact (Proof Lab v1.3 JSON)	7

1 Shadow Math: Core Definitions

Definition 1.1 (Shadow map). Let X be a latent space (e.g., $\mathbb{R}^D$, a manifold, or a metric space) and Y an observation space. A *shadow map* is a function $\pi : X \rightarrow Y$ that is typically surjective and not injective.

Definition 1.2 (Fiber / ambiguity set). For $y \in Y$, the *fiber* is $\pi^{-1}(y) = \{x \in X : \pi(x) = y\}$. When π is not injective, fibers contain multiple points: reconstruction from a single shadow is ill-posed without extra constraints.

Definition 1.3 (Lift). A *lift* is a (possibly partial) right-inverse $\ell : Y \rightarrow X$ such that $\pi(\ell(y)) = y$ whenever defined. In practice, lifts are induced by multi-shadow consistency, rigidity constraints, priors, or optimization.

Definition 1.4 (Multi-shadow system). A multi-shadow observation consists of maps $\pi_i : X \rightarrow Y_i$ for $i = 1, \dots, m$. A latent point $x \in X$ is observed as $(y_1, \dots, y_m)$ with $y_i = \pi_i(x)$. A *consistent lift* is an x such that $x \in \bigcap_{i=1}^m \pi_i^{-1}(y_i)$.

Remark 1.1 (What “higher dimension” means here). Throughout, “higher dimension” is *mathematical*: the minimal Euclidean dimension needed to realize the constraints. No physical claim is made about spacetime; we prove statements about geometric consistency and embedding dimension.

2 Theorem A: Single-View Non-Injectivity

2.1 Pinhole projection model

Let $\pi_f : \mathbb{R}_{z>0}^3 \rightarrow \mathbb{R}^2$ be calibrated pinhole projection with focal length $f > 0$:

$$\pi_f(X, Y, Z) = \left(\frac{fX}{Z}, \frac{fY}{Z} \right).$$

Theorem 2.1 (Ray fiber; no unique single-view lift). *For any $(u, v) \in \mathbb{R}^2$, the fiber $\pi_f^{-1}(u, v)$ contains infinitely many points (a 1D ray). Hence π_f is not injective.*

Proof. Fix (u, v) . The constraints $u = fX/Z$ and $v = fY/Z$ imply $X = (u/f)Z$ and $Y = (v/f)Z$ for any $Z > 0$. Therefore the entire ray $\{((u/f)Z, (v/f)Z, Z) : Z > 0\}$ maps to (u, v) , so the fiber is infinite and π_f is not injective. $\square$

Corollary 2.1 (Non-uniqueness of reconstruction from one image). *Any algorithm that reconstructs 3D from a single 2D image must inject extra assumptions (priors, learned models, symmetry, rigidity, known scale, etc.).*

3 Theorem B: Multi-View Lift via Triangulation

3.1 Back-projected rays

A calibrated camera defines $\pi_i : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with optical center C_i . For an observation $x_i \in \mathbb{R}^2$, the set of 3D points that project to x_i is a ray (or line) in 3D:

$$L_i(x_i) = \{C_i + t d_i(x_i) : t > 0\},$$

where $d_i(x_i)$ is the unit direction associated with pixel x_i .

Lemma 3.1 (Two distinct lines intersect in at most one point). *If L_1 and L_2 are distinct lines in $\mathbb{R}^3$, then $|L_1 \cap L_2| \leq 1$.*

Proof. If two distinct lines intersect at two points, they must coincide (a line is determined by two points), contradiction. $\square$

Theorem 3.1 (Two-view triangulation (noise-free, correct correspondence)). *Let cameras 1 and 2 be calibrated with distinct optical centers $C_1 \neq C_2$. Let (x_1, x_2) be a correspondence generated by a single finite 3D point X . Then X is uniquely determined and equals the unique intersection:*

$$X = L_1(x_1) \cap L_2(x_2).$$

Proof. Since (x_1, x_2) comes from X , we have $X \in L_1(x_1)$ and $X \in L_2(x_2)$ so the intersection is nonempty. Because $C_1 \neq C_2$, the rays are distinct lines. By Lemma 3.1, their intersection contains at most one point. Therefore the intersection is exactly $\{X\}$. $\square$

Remark 3.1 (Noise and practical triangulation). With noise, rays do not intersect exactly; one solves a least-squares approximation (e.g., DLT / SVD). Three or more views improve stability by over-determining the system.

4 Shadow Calculus: Fiber Dimension from Jacobian Rank

Theorem 4.1 (Constant Rank Theorem (informal statement)). *Let $\pi : \mathbb{R}^D \rightarrow \mathbb{R}^d$ be C^1 and suppose $\text{rank } D\pi(x) = r$ is constant in a neighborhood of x . Then near x , the fiber $\pi^{-1}(\pi(x))$ is a $(D - r)$ -dimensional submanifold.*

Remark 4.1. For a generic smooth projection from D to d with full rank $r = d$, the fiber is locally $(D - d)$ -dimensional. For pinhole projection $\mathbb{R}^3 \rightarrow \mathbb{R}^2$, this predicts a 1D ambiguity set (the ray), matching Theorem 2.1.

5 Theorem C: Gram Eigenstructure Certifies Dimension

5.1 Distances, centering, and Gram

Given points $p_1, \dots, p_n \in \mathbb{R}^k$, define squared distances $D_{ij}^{(2)} = \|p_i - p_j\|_2^2$ and the centering matrix $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. Define the double-centered Gram candidate

$$B = -\frac{1}{2}JD^{(2)}J.$$

Theorem 5.1 (Euclidean distance matrix equivalence). *If $D^{(2)}$ is generated by some points in $\mathbb{R}^k$, then $B \succeq 0$ and $\text{rank}(B) \leq k$. Conversely, if $B \succeq 0$ and $\text{rank}(B) = r$, then there exist points in $\mathbb{R}^r$ whose squared distances equal $D^{(2)}$.*

Proof. (Forward.) Let P be the centered coordinate matrix (rows p_i with mean zero). Then $G = PP^\top$ is positive semidefinite and $\text{rank}(G) \leq k$. Classical MDS shows $G = -\frac{1}{2}JD^{(2)}J = B$.

(Reverse.) If $B = U\Lambda U^\top$ with $\Lambda \succeq 0$, define $X = U_r\Lambda_r^{1/2}$ using the positive eigenpairs. Then $XX^\top = B$ is a Gram matrix in $\mathbb{R}^r$ and reproduces $D^{(2)}$. $\square$

Corollary 5.1 (Certified lower bound on embedding dimension). *If B has at least $k+1$ strictly positive eigenvalues, then the distances cannot embed isometrically in $\mathbb{R}^k$. Thus the intrinsic Euclidean dimension is at least $k+1$.*

5.2 Noise-robust planarity certificate

Let eigenvalues be $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \dots$ and define $\lambda_i^+ = \max(\lambda_i, 0)$.

Definition 5.1 (Out-of-plane ratios).

$$\rho := \frac{\lambda_3^+}{\lambda_2^+} \quad (\lambda_2^+ > 0), \quad \eta := \frac{\lambda_3^+}{\sum_i \lambda_i^+}.$$

Definition 5.2 (Energy effective dimension). Given $\varepsilon \in (0, 1)$,

$$k_{\text{eff}}(\varepsilon) = \min \left\{ k : \frac{\sum_{i>k} \lambda_i^+}{\sum_i \lambda_i^+} \leq \varepsilon \right\}.$$

Definition 5.3 (MDS stress (squared)). Let $\hat{X}^{(k)}$ be the classical MDS embedding using the top k eigenpairs of B . Let D be original distances and $\hat{D}^{(k)}$ be distances in $\hat{X}^{(k)}$. Define

$$\text{stress}(k) = \left(\frac{\|D - \hat{D}^{(k)}\|_F}{\|D\|_F} \right)^2.$$

Proposition 5.1 (Planar data implies $\lambda_3 = 0$ (noise-free)). *If points lie in an affine 2D subspace, then $\text{rank}(B) \leq 2$ and $\lambda_3 = 0$, so $\rho = \eta = 0$, $k_{\text{eff}} = 2$, and $\text{stress}(2) = 0$.*

Proof. After centering, points lie in a 2D linear subspace, so the centered coordinate matrix has rank at most 2. Thus $B = PP^\top$ has rank at most 2 and $\lambda_3 = 0$. Exact MDS in $k = 2$ reproduces all distances so stress is 0. $\square$

Definition 5.4 (Audit-grade planarity rule). Fix thresholds (ρ_τ, η_τ) and declare *planar* iff $\rho \leq \rho_\tau$ and $\eta \leq \eta_\tau$. In Proof Lab v1.3 we use $(\rho_\tau, \eta_\tau) = (0.01, 0.005)$.

6 Empirical Evidence: Proof Lab v1.3 (Audit Embed)

6.1 Experiment summary

The simulation uses: seed 7, $n = 120$, noise $\sigma = 0.7$ px, 3 views, focal $f = 800$ px, baseline 0.7, and certificate settings `energy_eps = 0.002`, thresholds $(\rho_\tau, \eta_\tau) = (0.01, 0.005)$.

6.2 Key outcomes (v1.3)

- **(A) Single-view non-injectivity witnessed:** pixel distance between two distinct 3D points sharing the same 2D projection is 0.0.
- **Reprojection sanity:** 3D reconstructed mean/median/max reprojection is 0.633/0.595/1.746 px; 2D control reconstructed is 0.624/0.572/1.759 px.
- **Dimension certificate:** 3D reconstructed is non-planar with $\rho = 0.458569$ and $\eta = 0.0857872$; $k_{\text{eff}} = 3$. 2D control reconstructed is planar with $\rho = 0.00314854$ and $\eta = 0.00114077$; $k_{\text{eff}} = 2$.
- **Ablation validation:** projecting the reconstructed 2D control to its best-fit plane collapses out-of-plane statistics: $\rho \approx 5.45439 \times 10^{-15}$, $\eta \approx 1.97848 \times 10^{-15}$, and $\text{stress}(2) \approx 6.03971 \times 10^{-15}$.

6.3 Audit table

Dataset	$\rho = \lambda_3/\lambda_2$	$\eta = \lambda_3/\sum \lambda_i$	k_{eff}	$\text{stress}(2)$
3D reconstructed	0.458569	0.0857872	3	0.0966937
2D reconstructed	0.00314854	0.00114077	2	0.00235905
2D recon $\rightarrow$ plane	5.45439×10^{-15}	1.97848×10^{-15}	2	6.03971×10^{-15}

Table 1: Noise-robust certificate statistics reported by Proof Lab v1.3.

7 Interpretation: What is Proven and What is Not

7.1 What is proven (mathematically)

- A single calibrated 2D pinhole projection is not injective; its fiber is a ray (Theorem 2.1).
- Two calibrated views with non-zero baseline uniquely recover a 3D point in the noise-free consistent case (Theorem 3.1).
- The Gram eigenstructure of distances certifies intrinsic Euclidean embedding dimension (Theorem 5.1 and Corollary 5.1).
- In noisy data, eigen-ratios (ρ), out-of-plane energy (η), and stress provide an audit-grade planarity classifier, validated by ablation (Proof Lab v1.3).

7.2 What is not proven (physics)

This document does not prove physical extra spacetime dimensions. It proves a rigorous statement of *dimensional necessity for constraints*: if a set of constraints (e.g., distances) is incompatible with any embedding in $\mathbb{R}^d$, then a higher-dimensional mathematical model is required.

A Appendix A: Embedded Audit Artifact (Proof Lab v1.3 JSON)

The full Proof Lab v1.3 report is embedded verbatim below (so the PDF is self-contained and auditable).

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